



Robust Underwater Direction-of-Arrival Tracking with Uncertain Measurement Noise

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Abstract. A robust underwater direction-of-arrival (DOA) tracking technique is presented by the paper. Firstly, considering the port and starboard ambiguity problem, the measurement model is founded as a non-linear sound pressure model based on a uniform circular array. Then, considering the unknown underwater environment can sometimes cause uncertain measurement noise, a robust DOA tracking technique based on variational Bayesian (VB) and extended Kalman filter (EKF) method is proposed. By utilizing proposed VB-EKF to estimate uncertain measurement noise online, DOA tracking technique can achieve a reliable tracking accuracy. Finally, the proposed VB-EKF is tested by simulations to verify its robustness and tracking performance.

Keywords: DOA tracking · Variational Bayesian · Extended Kalman filter

1 Introduction

Bearing angle estimation and tracking are important research areas in the field of aerospace engineering, remote sensing and tracking underwater target. For underwater target tracking scenario, passive bearing angle estimation can not only supply angle information of interested target, but also localize target with two or more different bearing angle information [1]. Usually, the bearing angle estimation techniques can be divided by two categories, namely the DOA estimation and the DOA tracking. The former one unusually assumes the target remains stationary or make very slow maneuvers. However, in many underwater target tracking scenes, the target always performs a typical moving mode (i.e., constant-velocity (CV) moving, constant-acceleration moving, etc.) that the DOA estimation technique can have poor estimation results since it ignores the target's kinematic and dynamic information [1]. Under this circumstance, the DOA tracking technique which considers both of the target's kinematics and dynamics and the measurement model can achieve more accurate estimation results if the target is maneuvering.

Considering the superior characteristics of the DOA tracking. Many researchers have proposed a number of nonlinear tracking algorithms to realize the DOA tracking in the scenario of underwater target bearing angle estimation. Kong and Chun [1] proposed

a novel extended Kalman filter (EKF) based DOA tracking method dealing with the nonlinear measurement model, which linearize the nonlinear model by applying first-order Taylor series expansion, i.e., Jacobian matrix, of measurement model. Depend on similar principles that linearize nonlinear models, high-precision tracking techniques using unscented Kalman filter [2] and cubature Kalman filter (CKF) [3, 4] are proposed. Unlike the above methods that approximately linearize nonlinear models, the tracking method based on particle filter (PF) [5] can theoretically realize accurate tracking in any tracking models with any distributions of stochastic noise by generating and transforming a large number of arbitrary particles. However, the PF-based tracking method has significant computational complexity due to its large number of particles, especially when tracking system is nonlinear. Then, real applications are very few based on the PF tracking method.

Also, taking account unknown underwater environment can sometimes lead to uncertain measurement noise, the DOA tracking technique needs to be robust to obtain accurate bearing angle estimations. The state-of-art robust tracking techniques can be divided to two categories, namely the inspired technique (i.e., Sage-Husa online noise estimator and innovation adaptive estimator) [6] and the Bayesian technique (i.e., variational Bayesian estimator) [7, 8]. The former one depends on a large period of good measurements and estimate the uncertain measurement covariance matrix by the historical good estimations. However, this kind of inspired online noise estimator can hardly guarantee its convergence and the period of good estimations is usually determined by experience. Considering the drawbacks of the inspired techniques, the VB technique is much more superior for it has the analytical description of the posterior probability function (PDF) of the uncertain measurement noise. Also, by adopting the VB estimation technique, both of the uncertain measurement noise and the current state can be estimated in the same time [7, 8]. As a result, the VB based online noise estimating techniques are more and more popular among the researchers to realize the robust target tracking missions.

Inspired by the VB technique and focused on the robust underwater target tracking mission, this paper proposed a VB-EKF to achieve robust and accurate DOA tracking of an underwater target with uncertain measurement noise. Firstly, considering a wide range in a long-time tracking scenario, the uniform circular array (UCA) is used since it is free from the port and starboard ambiguity problem [9]. Then, the VB-EKF is derived based on the high nonlinear measurement model and the uncertain measurement noise. Finally, the proposed VB-EKF is verified by the MATLAB simulations and the robustness and accuracy are proved.

2 Problem Formulation

2.1 Underwater Target's Bearing Angle Kinematic Model

Since underwater target often perform in a constant velocity to keep invisibility and far from sonar arrays, the kinematic model of the target's bearing angle can be modeled as the CV model.

Let θ_k denotes the bearing angle of the target (θ_k is angle between the target and the positive x-axis with the positive counterclockwise direction.) and $\dot{\theta}_k$ denotes the change

rate of θ_k . Then, the CV motion kinematic model of the bearing angle:

$$\mathbf{x}_k = \mathbf{F}_{k|k-1}\mathbf{x}_{k-1} + \mathbf{G}_{k-1}w_{k-1} \quad (1)$$

where $\mathbf{x}_k = (\theta_k, \dot{\theta}_k)^T$ is the system state at time step k , and $(\cdot)^T$ denotes the matrix transposition, and w_{k-1} denotes the zero mean Gaussian process noise with the covariance matrix Q_{k-1} caused by the unknown underwater environment. Q_{k-1} is named process noise covariance matrix (PNCM). $\mathbf{F}_{k|k-1}$ and $\mathbf{G}_k$ are the state transition matrix for CV motion and the noise driving matrix at time step k , which are given by:

$$\mathbf{F}_{k|k-1} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \quad (2)$$

$$\mathbf{G}_{k-1} = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} \quad (3)$$

where T is interval between the adjacent time steps.

2.2 Measurement Model Using UCAs

Considering the UCAs can avoid the port and starboard ambiguity problem, the measurement model is founded by the UCAs raw measurement. The configuration of UCAs and the underwater target is depicted as Fig. 1.

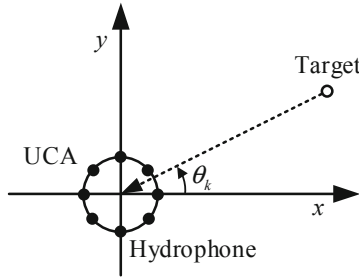


Fig. 1. Configuration of the UCA based measurement system

Considering the target emits a narrowband acoustic signal at the frequency f with the amplitude a and the initial phase ϕ_0 , the target signal at k is expressed as:

$$s(k) = \alpha \exp(j(2\pi fk + \phi_0)) \quad (4)$$

The emitted signal given by Eq. (4) is received by the UCA with P elements at time step k . The P elements are uniformly placed around the circle with the radius and the center at origin of the cartesian coordinates. It is assumed that the range of the target satisfies the far-field condition to regard the arrived signal as a plane wave, and the sound velocity is c . Then, time delay of signal relative to center of UCA at the p -th element

($p = 1, 2, \dots, P$) is $r/c \cos(\theta_k - 2\pi p/P)$. Also, considering the signal given by Eq. (4), the shift of the signal caused by the time delay at the p -th element is expressed as:

$$A_p(\theta_k) = \exp\left(j2\pi f \frac{r}{c} \cos\left(\theta_k - 2\pi \frac{p}{P}\right)\right) \quad (5)$$

Furthermore, the received real signal of the p -th element $z_p(k)$ is expressed as:

$$z_p(k) = \text{real}(A_p(\theta_k)s(k)) + v_k \quad (6)$$

where $\text{real}(\cdot)$ denotes taking real part of a complex number, θ_k is bearing angle of the target at time k , v_k is the measurement noise at step k , which is defined as the zero mean Gaussian noise with the variance $\sigma_{r,k}^2$ from the unknown underwater environment and the measurement system.

Defining the received signal vector of UCA at step k as $\mathbf{z}(k) = [z_1(k), z_2(k), \dots, z_P(k)]^T$, the output of the UCA can be expressed as:

$$\mathbf{z}(k) = \text{real}(\mathbf{A}(\theta_k)s(k)) + \mathbf{v}_k \quad (7)$$

where $\mathbf{h}_k(\mathbf{x}_k)$ denotes the nonlinear measurement function, and $\mathbf{A}(\theta_k) = [A_1(\theta_k), A_2(\theta_k), \dots, A_P(\theta_k)]^T$ is the array manifold vector of the UCA, and $\mathbf{v}_k$ is the column vector formed by the measurement noise of P elements. M snapshots of the received signal are taken in a measurement. The snapshots are taken from a small period within T . Then, the measurement model at tracking step k is expressed as:

$$\mathbf{Z}_k = \begin{bmatrix} \mathbf{z}(k) \\ \mathbf{z}(k + \tau) \\ \mathbf{z}(k + 2\tau) \\ \vdots \\ \mathbf{z}(k + (M - 1)\tau) \end{bmatrix} = \mathbf{h}_k(\mathbf{X}_k) + \mathbf{V}_k = \begin{bmatrix} \text{real}(\mathbf{A}(\theta_k)s(k)) \\ \text{real}(\mathbf{A}(\theta_k)s(k + \tau)) \\ \text{real}(\mathbf{A}(\theta_k)s(k + 2\tau)) \\ \vdots \\ \text{real}(\mathbf{A}(\theta_k)s(k + (M - 1)\tau)) \end{bmatrix} + \mathbf{V}_k \quad (8)$$

where τ is the interval between adjacent snapshots, and $M\tau \ll T$. It is assumed that the stochastic measurement noise of each element is independent and subject to the same distributions. Thus, the measurement noise $\mathbf{V}_k$ at tracking step k is defined as a zero-mean Gaussian noise with MNCM of $\mathbf{R}_k = \mathbf{I}_{MP}\sigma_{r,k}^2$, where $\mathbf{I}_{MP}$ is the MP -order identity matrix and $\sigma_{r,k}^2$ is the variance of measurement noise of each element at tracking step k .

From Eq. (8), it can be found that the measurement model is high nonlinear. Also, the unknown underwater environment sometimes will change the MNCM to make it uncertain. Thus, based on the VB technique, the VB-EKF DOA tracking method is derived in detail in this section.

3 VB-EKF for Robust DOA Tracking

3.1 Choice of Prior Distribution

In order to infer $\mathbf{x}_k$ along with $\mathbf{R}_k$, a conjugate prior need to select for the fluctuant MNCM $\mathbf{R}_k$. Since a conjugate distribution can guarantee the same functional forms of distribution

and the posterior distribution. In the Bayesian theory, inverse Wishart distribution is usually used as the conjugate prior for the covariance matrix of a Gaussian distribution with known mean [10]. Since $\mathbf{R}_k$ is the covariance matrix of Gaussian distribution, the prior distribution $p(\mathbf{R}_k | \mathbf{z}_{1:k-1})$ is selected as an inverse Wishart distribution given by

$$p(\mathbf{R}_k | \mathbf{z}_{1:k-1}) = \text{IW}(\mathbf{R}_k; \hat{u}_{k|k-1}, \hat{U}_{k|k-1}) \quad (9)$$

where $\text{IW}(\cdot; \lambda, \Psi)$ denotes the PDF of the inverse Wishart distribution with degree of freedom (dof) λ and inverse scale matrix Ψ [8], $\hat{u}_{k|k-1}$ and $\hat{U}_{k|k-1}$ are the dof and the inverse scale matrix of $p(\mathbf{R}_k | \mathbf{z}_{1:k-1})$, respectively.

The posterior distribution $p(\mathbf{R}_{k-1} | \mathbf{z}_{1:k-1})$ is also subject to an inverse Wishart distribution as follows:

$$p(\mathbf{R}_{k-1} | \mathbf{z}_{1:k-1}) = \text{IW}(\mathbf{R}_{k-1}; \hat{u}_{k-1|k-1}, \hat{U}_{k-1|k-1}) \quad (10)$$

To guarantee that $p(\mathbf{R}_k | \mathbf{z}_{1:k-1})$ is the inverse Wishart distribution given by Eq. (9), the previous approximate posteriors is spread through a forgetting factor $\rho \in (0, 1]$, which indicates the extend of the time-fluctuations of the MNMCM. Then, the prior dof $\hat{u}_{k|k-1}$ and the prior inverse scale matrix $\hat{U}_{k|k-1}$ are given as follows [8]:

$$\hat{u}_{k|k-1} = \rho(\hat{u}_{k-1|k-1} - n - 1) + n + 1 \quad (11)$$

$$\hat{U}_{k|k-1} = \rho \hat{U}_{k-1|k-1} \quad (12)$$

where n denotes the order of the MNMCM $\mathbf{R}_k$.

3.2 VB-EKF Algorithm

Combining the of framework of the standard EKF and the assumption of the distribution of MNMCM, the algorithm of VB-EKF at one time step is given as follows:

1) Input parameters:

Input the estimate $\hat{x}_{k-1|k-1}$ and root mean square matrix $\mathbf{P}_{k-1|k-1}$ at time $k-1$, the dof $\hat{u}_{k-1|k-1}$ and the inverse scale matrix $\hat{U}_{k-1|k-1}$ at step $k-1$, measurement $\mathbf{z}_k$ at time step k , the forgetting factor ρ , and number of iterations N .

2) Time update:

One step prediction state and root mean square error are given:

$$\hat{x}_{k|k-1} = \mathbf{F}_{k-1} \hat{x}_{k-1|k-1} \quad (13)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k|k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k|k-1}^T + \mathbf{G}_{k-1} \mathbf{Q}_{k-1} \mathbf{G}_{k-1}^T \quad (14)$$

The dof and inverse scale matrix are given by:

$$\hat{u}_{k|k-1} = \rho(\hat{u}_{k-1|k-1} - n - 1) + n + 1 \quad (15)$$

$$\hat{U}_{k|k-1} = \rho \hat{U}_{k-1|k-1} \quad (16)$$

3) Iteration process of measurement update:

The state and the estimate of MNCM is initialized as follows:

$$\hat{x}_k^{(0)} = \hat{x}_{k|k-1} \quad (17)$$

$$\hat{R}_k^{(0)} = \hat{U}_{k|k-1} / \hat{u}_{k|k-1} \quad (18)$$

Let $i = 0$ and start the iteration process.

The estimate of the signal of the target is given by [1]:

$$\hat{s}_{k|k}^{(i)}(k) = \left(\mathbf{A}^H(\hat{\theta}_{k|k}^{(i)}) \mathbf{A}(\hat{\theta}_{k|k}^{(i)}) \right)^{-1} \mathbf{A}^H(\hat{\theta}_{k|k}^{(i)}) \text{Hilbert}(z(k)) \quad (19)$$

where $\hat{\theta}_{k|k}^{(i)}$ is the predicted bearing angle, i.e., the first term of $\hat{x}_{k|k-1}$, $\mathbf{A}(\hat{\theta}_{k|k}^{(i)})$ is obtained by using Eq. (5). $(\cdot)^H$ and $\text{Hilbert}(\cdot)$ denote the conjugate transposition and the Hilbert transform, respectively.

The measurement function $h(\hat{x}_{k|k}^{(i)})$ is obtained by substituting $\hat{x}_{k|k}^{(i)}$ into Eq. (8). The Jacobian matrix of the measurement function is given by:

$$\mathbf{H}_k^{(i)} = \begin{bmatrix} h'_1, h'_2 \cdots h'_p \\ 0, 0, \cdots, 0 \end{bmatrix}^T \quad (20)$$

where the expression of h'_p is given as follows:

$$h'_p = -2\pi f \frac{r}{c} \sin\left(\hat{\theta}_{k|k}^{(i)} - 2\pi \frac{p}{P}\right) \text{real}\left(jA_p\left(\hat{\theta}_{k|k}^{(i)}\right)\hat{s}_{k|k-1}(k)\right) \quad (21)$$

The Kalman filter gain is updated as:

$$\mathbf{K}_k^{(i+1)} = \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T (\mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{R}_k^{(i)})^{-1} \quad (22)$$

The state is updated as:

$$\hat{x}_{k|k}^{(i+1)} = \hat{x}_{k|k-1} + \mathbf{K}_k^{(i+1)} (z_k - h(\hat{x}_{k|k}^{(i)}) - \mathbf{H}_k^{(i)} (\hat{x}_{k|k-1} - \hat{x}_{k|k}^{(i)})) \quad (23)$$

The root mean square error matrix is updated as:

$$\mathbf{P}_{k|k}^{(i+1)} = \mathbf{P}_{k|k-1} - \mathbf{K}_k^{(i+1)} \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} \quad (24)$$

$\hat{s}_{k|k}^{(i+1)}(k)$, $h(\hat{x}_{k|k}^{(i+1)})$, and $\mathbf{H}_k^{(i+1)}$ are calculated by Eq. (8), (19)-(21).

The dof and the inverse scale are updated as follows:

$$\mathbf{B}_k^{(i+1)} = (z_k - h(\hat{x}_{k|k}^{(i+1)}))(z_k - h(\hat{x}_{k|k}^{(i+1)}))^T + \mathbf{H}_k^{(i+1)} \mathbf{P}_{k|k}^{(i+1)} (\mathbf{H}_k^{(i+1)})^T \quad (25)$$

$$\hat{U}_{k|k}^{(i+1)} = \hat{U}_{k|k-1} + \mathbf{B}_k^{(i+1)}, \quad \hat{u}_{k|k}^{(i+1)} = \hat{u}_{k|k-1} + 1 \quad (26)$$

The estimate of MNCM is updated as:

$$\hat{R}_k^{(i+1)} = \hat{U}_{k|k}^{(i+1)} / \hat{u}_{k|k}^{(i+1)} \quad (27)$$

Let $i = i + 1$ and perform step (2)-(10) until $i = N - 1$. Then the estimate of MNCM, the dof and the inverse scale matrix are decided to $\hat{R}_k = \hat{R}_k^{(N)}$, $\hat{u}_{k|k} = \hat{u}_{k|k}^{(N)}$, $\hat{U}_{k|k} = \hat{U}_{k|k}^{(N)}$.

4) Output parameters:

Output the state estimate $\hat{x}_{k|k}$, the root mean square error matrix $\mathbf{P}_{k|k}$, the estimates of dof $\hat{u}_{k|k}$ and inverse scale matrix $\hat{U}_{k|k}$.

By substituting the output parameters of the previous time step into the next time step as the input parameter, the state of the target can be estimated recursively.

4 Simulations

4.1 Simulation Scenario

Through simulations, an underwater target perform in the CV motion mode with the condition $\mathbf{x}_0 = [270, -0.03]$, and $\mathbf{Q}_{k-1} = 2 \times 10^{-4} \text{ s}^2$. The total simulation time was set to 5000 s with a 2 s interval. For the measurement model, the frequency f , amplitude a and the initial phase ϕ_0 of the target signal were 170 Hz, 1 and 0, respectively, and the sample frequency of the received signal f_s was 1000 Hz. A 12 elements UCA was used to obtain the measurements. The underwater sound velocity c was assumed to be 1500 m/s.

To test the robustness of VB-EKF to uncertain measurement noise caused by the unknown underwater circumstance, a strong Gaussian white noise was added from 3600 s to 4000 s of the measurements. Once the MNCM $\mathbf{R}_k$ during the period without the strong noise was set to $\mathbf{I}_P \sigma_r^2$, $\mathbf{R}_k$ at all steps was expressed as:

$$\mathbf{R}_k = \begin{cases} \mathbf{I}_P \sigma_r^2 & 0 \leq k < 1800, 2000 < k \leq 2500 \\ 10\mathbf{I}_P \sigma_r^2 & 1800 \leq k \leq 2000 \end{cases} \quad (28)$$

where σ_r^2 was set to 1.

The proposed VB-EKF for DOA tracking given by Algorithm 1 was tested by the simulated measurement data. The results of the Conventional beamforming (CBF)-based DOA estimation, the standard EKF is also given for comparison. The scanning bearing angle of the CBF varying every 1° from 0 to 360° . The state estimate of EKF and VB-EKF are initialized to $\hat{\mathbf{x}}_0 = [\hat{\theta}_{CBF}^0, 0]$, where $\hat{\theta}_{CBF}^0$ was the bearing angle estimated by using CBF at $k = 0$. $\mathbf{P}_{k|k}$ was initialized to $\mathbf{P}_0 = \text{diag}([100, 0.01])$. The process noise covariance matrix $\mathbf{Q}_{k-1}$ for EKF and VB-EKF are all set to the true value $2 \times 10^{-4} \text{ s}^2$. The MNCM of EKF was set to a constant $\mathbf{I}_P$. For the VB-EKF and, the dof $\hat{u}_0$ and inverse scale matrix $\hat{U}_0$ of the inverse Wishart distribution is set to 16 and $16\mathbf{I}_{MP}$, respectively, and the forgetting factor ρ was set to 0.9995.

To statistically evaluate performance of VB-EKF, 500 Monte Carlo simulations were performed. The root mean square error for the DOA tracking result at time step k was defined as

$$\text{RMSE}_\theta = \sqrt{\frac{1}{N} \sum_{n=1}^N (\hat{\theta}_k - \theta_k)^2} \quad (29)$$

where N is Monte Carlo trials' number, $\hat{\theta}_k$ is bearing angle estimate of the target, θ_k is true bearing angle of target.

4.2 Simulation Results

The DOA tracking results shown in Fig. 2, which showed that CBF was invalid during the existence of the strong noise from 3600 s to 4000 s due to its excessive relying on the present measurement. The precision of the EKF was higher than that of the CBF. Furthermore, the EKF provided valid tracking even from 3600 s to 4000 s. The VB-EKF provided the most high-precision tracking, even when the strong noise exists.

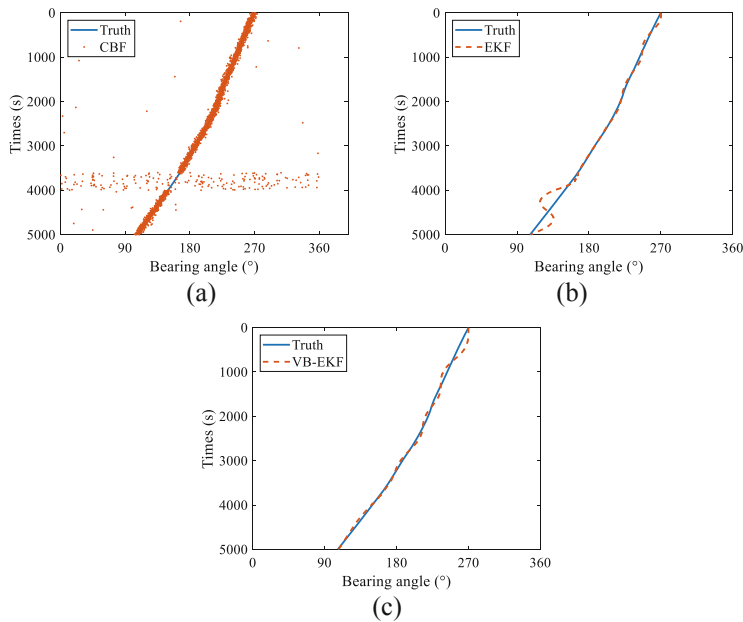


Fig. 2. Bearing angle tracking results of (a) CBF, (b) EKF, (c) VB-EKF

The accuracy of the proposed VB-EKF, EKF and CBF are shown in Fig. 3. It shows that, similar to the above conclusion, the VB-EKF provided the most high-precision tracking even during the existence of the strong noise, while the CBF was invalid during this time. Moreover, the performance of the VB-EKF was scarcely affected by the strong noise. The superior performance of the proposed VB-EKF was verified again.

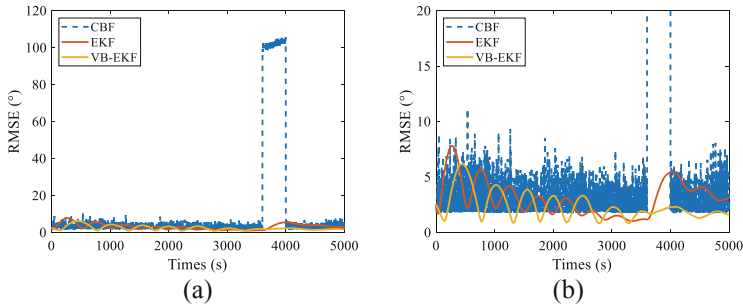


Fig. 3. RMSEs of CBF, EKF and VB-EKF.

5 Conclusion

A robust underwater target DOA tracking algorithm inspired of VB technique is proposed. By modeling measurement model of the raw measurement data of a UCA and considering the uncertain measurement noise, the proposed VB-EKF can make reliable estimations about the target's bearing angle. The simulation results proved the proposed VB-EKF has superior tracking performance compared to the CBF and EKF. Further work will focus on the computational effective variations of the robust DOA tracking based on VB and their experiment verification.

Acknowledgement. This work was supported by the National Natural Science Foundation of China under Grant No. 12104113, and the foundation of the State Key laboratory of AFDL under Grant No. 6142210200103.

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